MILITARY SPECIFICATION

TELESCOPE, ELBOW: M118 SERIES AND T176E4

This Amendment forms a part of Military Specification MIL-T-46316B(MU) dated 10 July 1968, and is approved for use by all Departments and Agencies of the Department of Defense.

Paragraph 1.1, delete and substitute as follows:

"1.1 This specification covers similar types of instruments designated as Telescope, Elbow: M118, M118C, M118CA1 and T176E4. These telescopes are basically identical except for differences in reticle patterns."

Paragraph 2.1: Under Drawings, insert: "M118CA1" after M118C.

Under Packaging Data Sheet, first line, delete: "og" and substitute "of".

Under Packaging Data Sheet, add: "10559855" and "Packaging of Telescope, Elbow: M118CA1".

Paragraph 3.3.1, delete and substitute as follows:

"3.3.1 Storage temperatures.- The telescope shall show no evidence of physical failure at room temperature (60° to 90°F) after having been thermally stabilized at -70°F and +160°F."

Paragraph 3.3.2, delete and substitute as follows:

"3.3.2 Standard ambient temperatures.- Subsequent to meeting the requirements of 3.3.1, the telescope shall meet all the requirements of this specification except 3.3.3 at a standard ambient temperature between 60°F and 90°F."

Paragraph 3.3.3, delete and substitute:

"3.3.3 Operating temperatures.- The telescope shall show no evidence of physical failure and shall meet the applicable requirements of 3.9 while thermally stabilized at -40°F and +150°F."

Paragraph 3.3.4, insert new paragraph as follows:

"3.3.4 Vibration.- The telescope shall show no evidence of physical failure (loose, missing or broken parts) after having been vibrated at a frequency of 30 cycles per second at an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes + 15 seconds. Vibration

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shall not have caused the eyepiece arm to disengage from the initially engaged position. Subsequent to vibration, the telescope shall meet the requirements of 3.4 to 3.10 inclusive.

Paragraph 3.5, delete and substitute as follows:

"3.5 Sealing.- The telescope shall maintain an internal pressure of dry nitrogen at 5 pounds per square inch (psig) for a minimum period of 2 hours without loss of pressure in excess of 0.2 psig. The interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F."

Paragraph 3.6, second line: Delete "kingpin" and substitute "tapered".

Paragraph 3.7.1, second line: Delete "the reticle" and substitute "both reticles."

Paragraph 3.7.5, delete and substitute as follows:

"3.7.5 Boresight error due to reticle rotation.- The line of sight at the boresight point shall not deviate more than 0.25 mil when the reticle is canted a minimum of 5 degrees to the left and a minimum of 5 degrees to the right by means of the cant correction knob."

Paragraph 3.9, fifth line: Delete "3.3.2" and substitute "3.3.3".

Paragraph 3.9.1, second line: Delete "kingpin" and substitute "tapered".

Paragraph 3.9.3, delete and substitute as follows:

"3.9.3 Mirror, plunger, arm and rheostat.- The level vial mirror (pivotal type only), plunger knob on the reticle shift lever, eyepiece arm and the rheostat knob shall be mechanically operable when exposed to the operating temperatures of 3.3.3."

Paragraph 4.3.2.1, delete from first sentence: "Listed in the list of inspection documents (IEL)."

Table I, change: "3.3.3" to "3.3.4".

Table II, change: "3.3.2" to "3.3.3".

Paragraph 4.6.1.1, change: "3.3.1 and 3.3.2" to "3.3.1, 3.3.2 and 3.3.3" in two places.

Paragraph 4.7.2.2, change: "3.3.3" to "3.3.4" in two places.

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* Paragraph 4.7.2.5, delete and substitute as follows:

"4.7.2.5 Eyepiece focus.- This test is to be accomplished by placing a white background in front of the telescope's objective lens. Eyepiece focus is to be measured using a 3 power dioptometer conforming to the accuracies depicted on Drawing 7680631. Adjust the dioptometer eyepiece assembly to the dioptometer's reticle until a clear sharp image appears on the dioptometer's optical axis. Next direct the dioptometer's line of sight toward a 500 feet distant target such as a smoke stack, edge of a building, or some other well defined target image. Adjust the objective assembly until parallax is removed. Set the dioptometer scale to zero diopters. The telescope reticle and image shall remain clear and distinguishable throughout the permissible diopter range specified in paragraph 3.7.6.

Paragraph 5.1, add the following: "10559855 Telescope Elbow M118CA1".

* The outside margins have been marked to indicate where additions to the previous amendment have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in this notation. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notation and relationship to the last previous amendment.

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